

Machine Learning Fundamentals

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What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence that enables systems to learn from data and improve performance without being explicitly programmed. ML algorithms discover patterns in data instead of following hand-coded rules. The goal is to build models that generalize well to unseen data.

Supervised Learning

The algorithm learns from labeled training data (input-output pairs). Classification predicts categories such as spam detection or image recognition. Regression predicts continuous values such as house prices or temperature. Common algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest, SVM, Neural Networks.

Unsupervised Learning

Finds patterns in unlabeled data. Clustering groups similar data points together: K-Means assigns each point to the nearest centroid, DBSCAN finds density-based clusters of arbitrary shape. Dimensionality Reduction: PCA reduces features by projecting data onto principal components. t-SNE is used for visualization.

Reinforcement Learning

An agent learns by interacting with an environment and receiving rewards or penalties. Used in game playing (AlphaGo, chess engines), robotics, autonomous driving, and recommendation systems. The agent learns a policy that maximizes cumulative reward over time.

The ML Pipeline

Step 1: Problem Definition - define what to predict. Step 2: Data Collection. Step 3: Data Preprocessing - handle missing values, outliers, scaling, encoding. Step 4: Exploratory Data Analysis. Step 5: Feature Engineering. Step 6: Model Selection and Training. Step 7: Evaluation on test set. Step 8: Hyperparameter Tuning. Step 9: Deployment.

Evaluation Metrics

Classification: Accuracy = $(TP+TN)/\text{Total}$. Precision = $TP/(TP+FP)$. Recall = $TP/(TP+FN)$. F1 = $2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$. ROC-AUC measures the area under the receiver operating characteristic curve. Regression: MSE, RMSE, MAE, R-squared (coefficient of determination).

Overfitting vs Underfitting

Overfitting occurs when a model memorizes training data but fails on new data. It is caused by excessive model complexity or insufficient training data. Solutions: regularization (L1/L2), dropout, cross-validation, data augmentation. Underfitting occurs when the model is too simple to capture patterns. Solutions: more complex model, better feature engineering, less regularization.